

Connexium Managed Switches - Redundancy Capabilities

Connexium Managed Switch

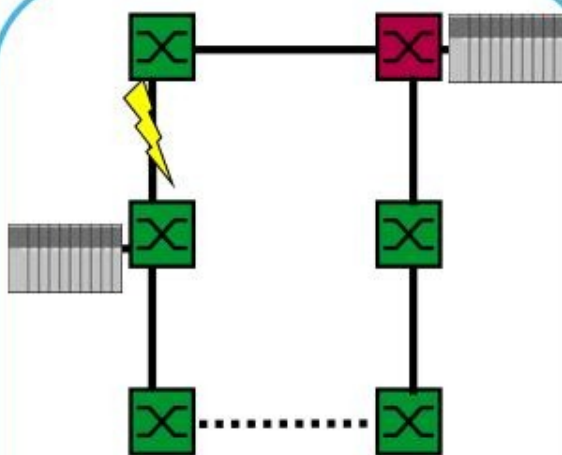
Redundancy

- The goal of network redundancy is to provide high network availability with automatic communication recovery in the event of a broken or unplugged cable or the shut down of a switch.
- The network recovery includes the time for the network to automatically reconfigure the communication path to all devices and reinitiate communications
- Redundant networks are typically configured in either a ring or mesh architecture depending on the selected redundancy protocol.
- All switches and devices on the network must support the same redundancy protocol
- The Connexium Managed Switches support several Redundancy Protocols that can be selected as part of the switch configuration process.

Redundancy Protocols

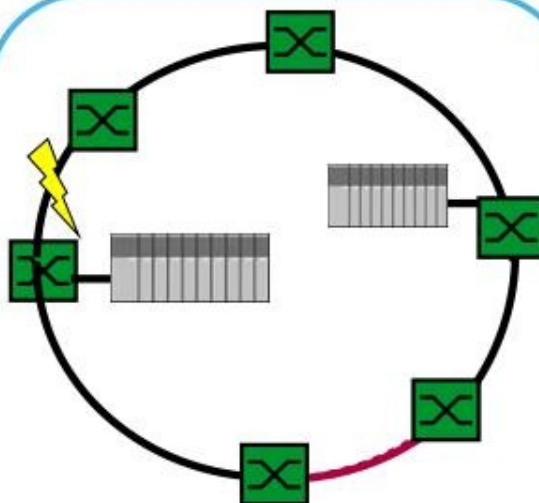
Connexium switches offer 3 different protocols to support network redundancy for High Availability

RSTP (Rapid Spanning Tree)



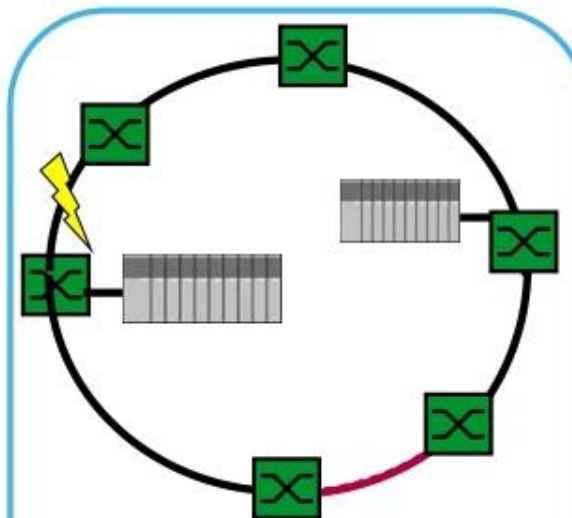
- RSTP compatible switches can be used.
- Multiple paths between devices are established.

HIPER-Ring



- Only Schneider switches
- Ability to couple rings.
- Fast HIPER-Ring option with Extended Switches

MRP



- MRP-compatible switch can be used.
- Ability to couple rings.

Connexium Managed Switch

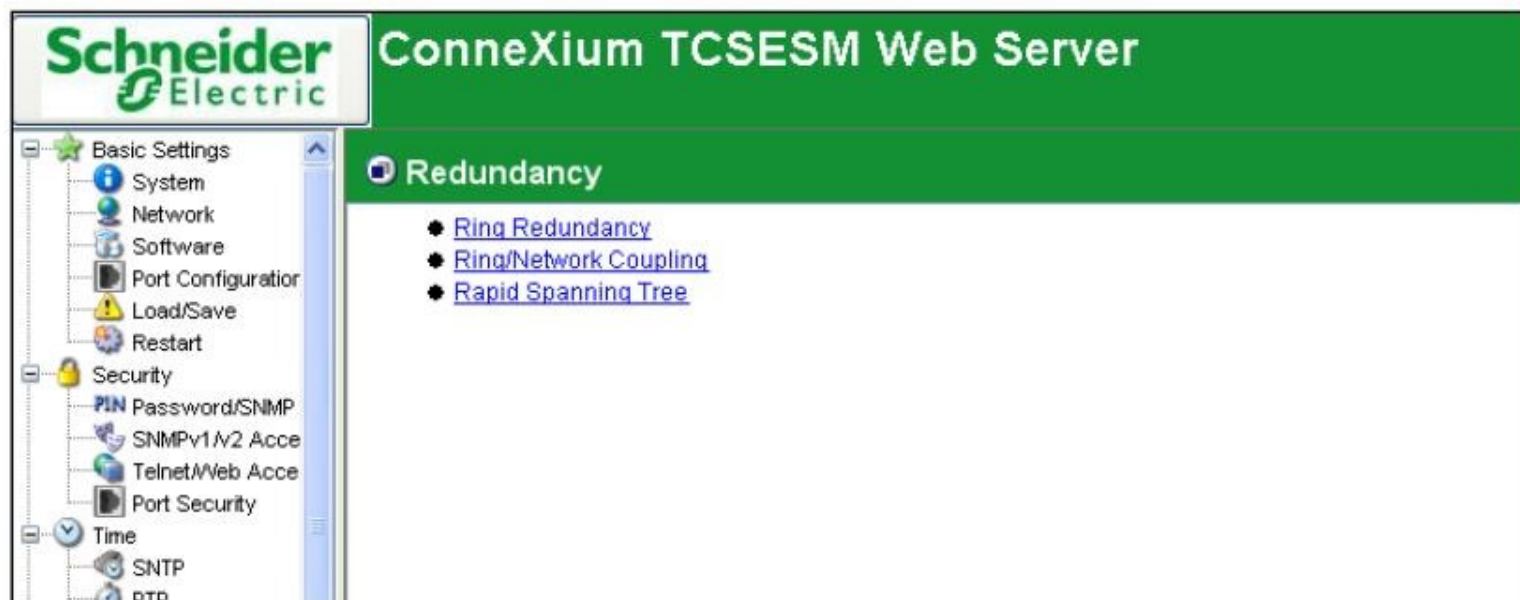
Redundancy Capabilities

- **RSTP (Rapid Spanning Tree Protocol)** - Industry standard protocol (IEEE 802.1) enabling deployment of ring and mesh architectures.
- **MRP (Media Redundancy Protocol)** - An IEC standard (IEC 62439) enabling deployment of ring architectures.
- **HIPER-Ring (High Performance Redundancy)** - A propriety protocol for ring configurations. **Fast HIPER-Ring** available with the extended switches
- **Ring Coupling** – Interconnecting 2 ring that are using either the MRP or HIPER-Ring protocols.
- **Dual Input Power Supply** - Dual power source connections to the switch power supply.

Redundancy Capabilities

Redundancy Capabilities Selection

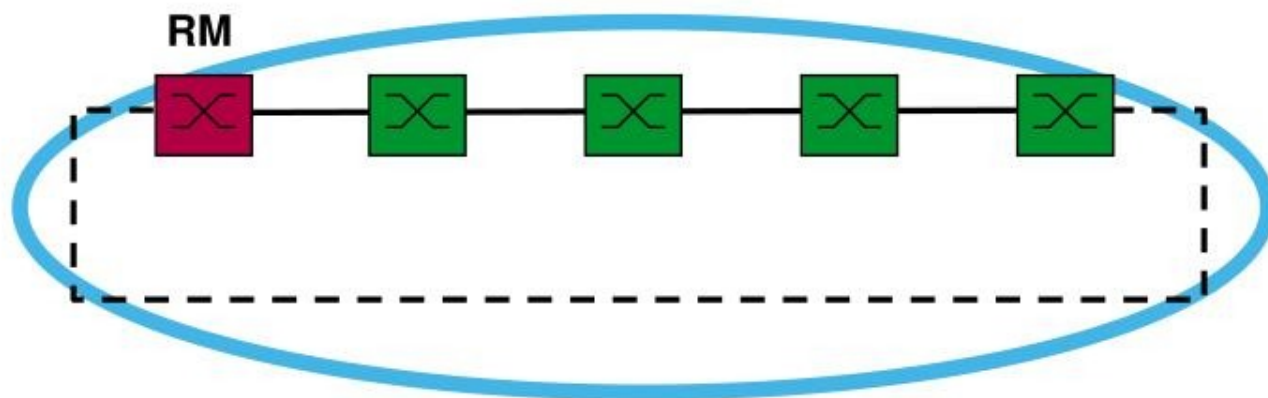
- The web page below allows the selection of the network redundancy capability that includes:
 - Ring Redundancy using the HIPER-Ring and MRP protocols
 - Ring/Network Coupling
 - Rapid Spanning Tree using RSTP protocol



Redundancy Capabilities

Ring/ Redundancy

- Ring redundancy, using HIPER-Ring or MRP Protocol, allows a single line structure network to be adapted to high-availability, ring-shaped network.



RM – Ring Manager, switch that manages the connection to the redundant line
— Main line
- - Redundant Line

Redundancy Capabilities

Ring Redundancy

- This web page allows the selection of:
 - Redundancy protocol, either HIPER-Ring or MRP protocols
 - Redundancy Manager mode
 - Ring Recovery, standard or accelerated mode

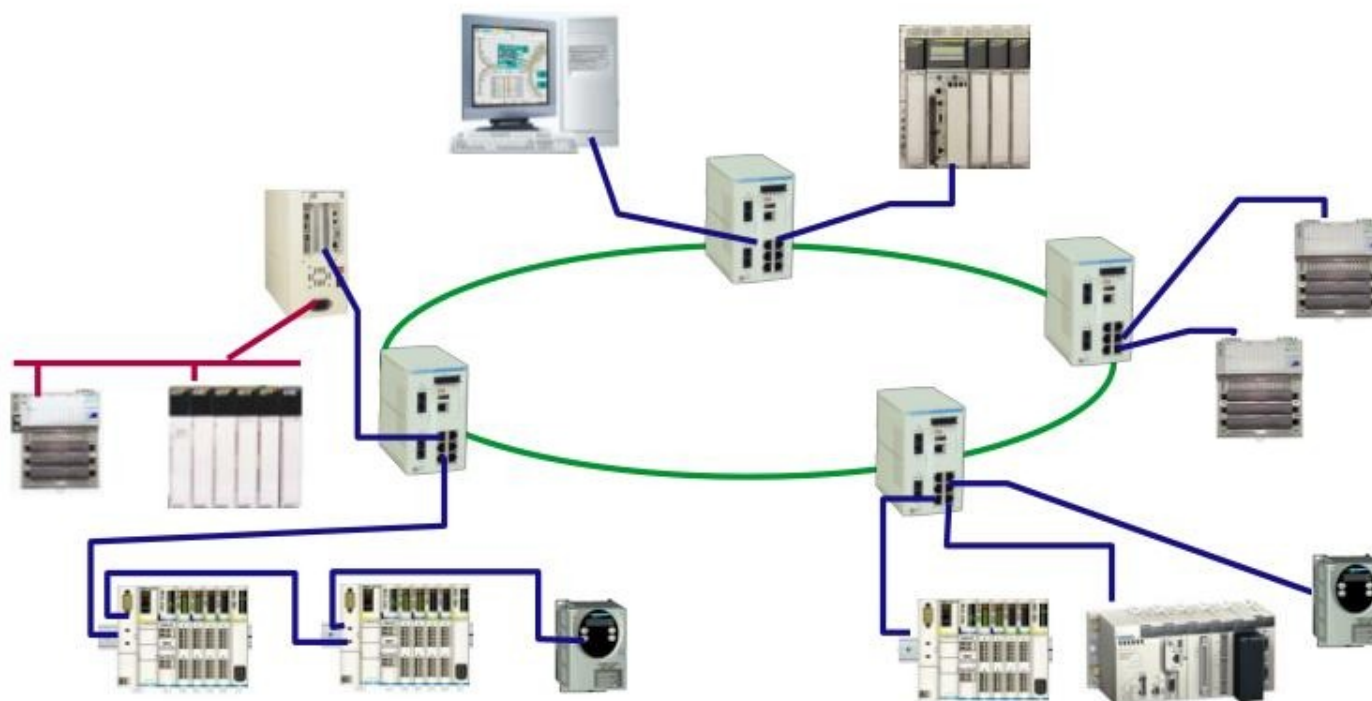
The screenshot shows a web interface for configuring Ring Redundancy. It features a green header bar with the title "Ring Redundancy". Below the header, the interface is organized into several sections:

- Version:** A section with two radio buttons: "HIPER-Ring" (selected) and "MRP".
- Ring Port 1:** A section with three input fields: "Module" (value: 1), "Port" (value: 1), and "Operation" (value: active).
- Ring Port 2:** A section with three input fields: "Module" (value: 1), "Port" (value: 2), and "Operation" (value: inactive).
- Redundancy Manager Status:** A section with two radio buttons: "Active (redundant line)" and "Inactive".
- Redundancy Manager:** A section with a "Mode" label and two radio buttons: "On" and "Off" (selected).
- Ring Recovery:** A section with two radio buttons: "Standard" (selected) and "Accelerated".
- Information:** A section with a single input field.

Redundancy

Single Ring Architecture

- Single Ring network, when cable break is detected, network recovers by automatically re-routed traffic



Ring Redundancy Protocol

Recovery Times

	HIPER-Ring	Fast HIPER-Ring	MRP
Recovery Time – ConneXium - Standard or Basic Switches	300 or 500 ms for up to 50 switches	NA	200 or 500 ms for up to 50 switches
Recovery Time – ConneXium Extended Switches	300 or 500 ms for up to 50 switches	25 ms for up to 50 switches	200 or 500 ms for up to 50 switches

Redundancy Capabilities

Ring/Network Coupling

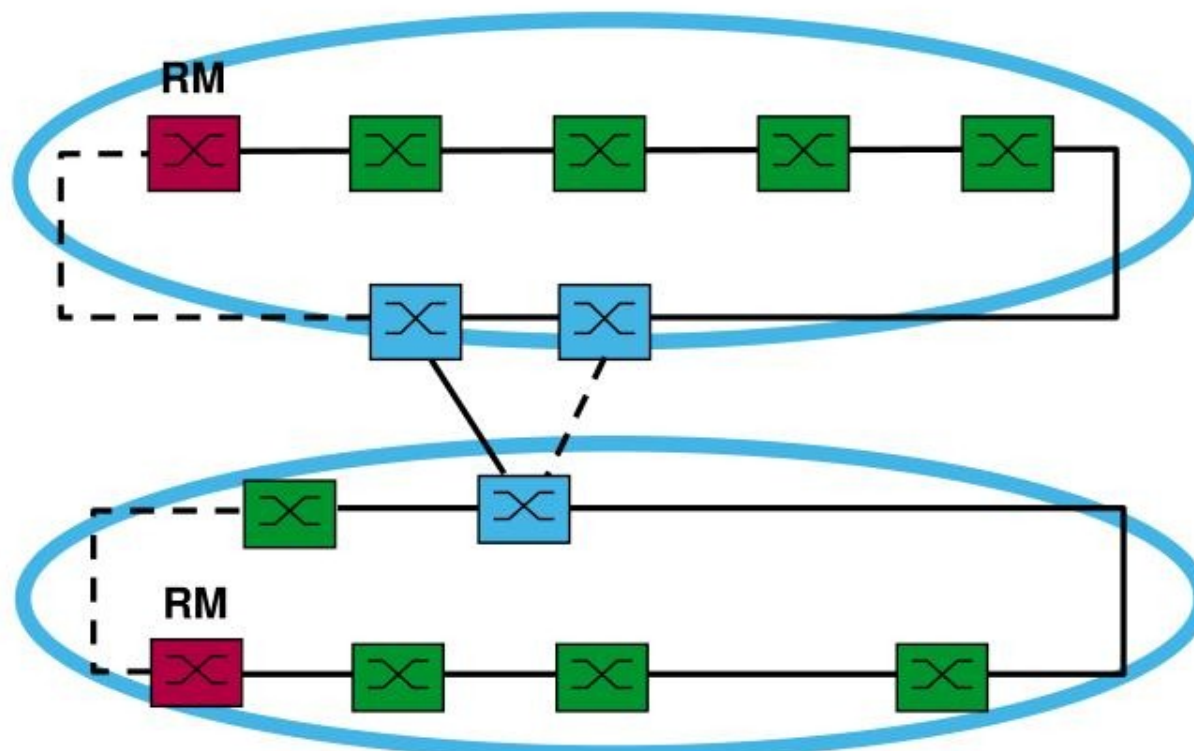
- Coupling allows the connection of 2 rings or network segments using single or redundant switches.
 - **Single switch coupling**, the redundant coupling is performed using 2 ports on a switch in the first ring/network, each port is connected to a port on 2 different switches in a second ring/network. One connection is the primary connection and the other is the stand-by connection.
 - **Redundant switch coupling**, the redundant coupling is performed by 1 port each on the two switches on the first ring/network to 1 port each of 2 switches on the second ring/network. One connection is the primary connection and the other is the stand-by connection.

Redundancy Capabilities

Ring/Network Coupling

- Single switch coupling network configuration

- Ring coupling detects main line inoperable, the redundant line resumes the communication typically within 150 ms (max. 500 ms)



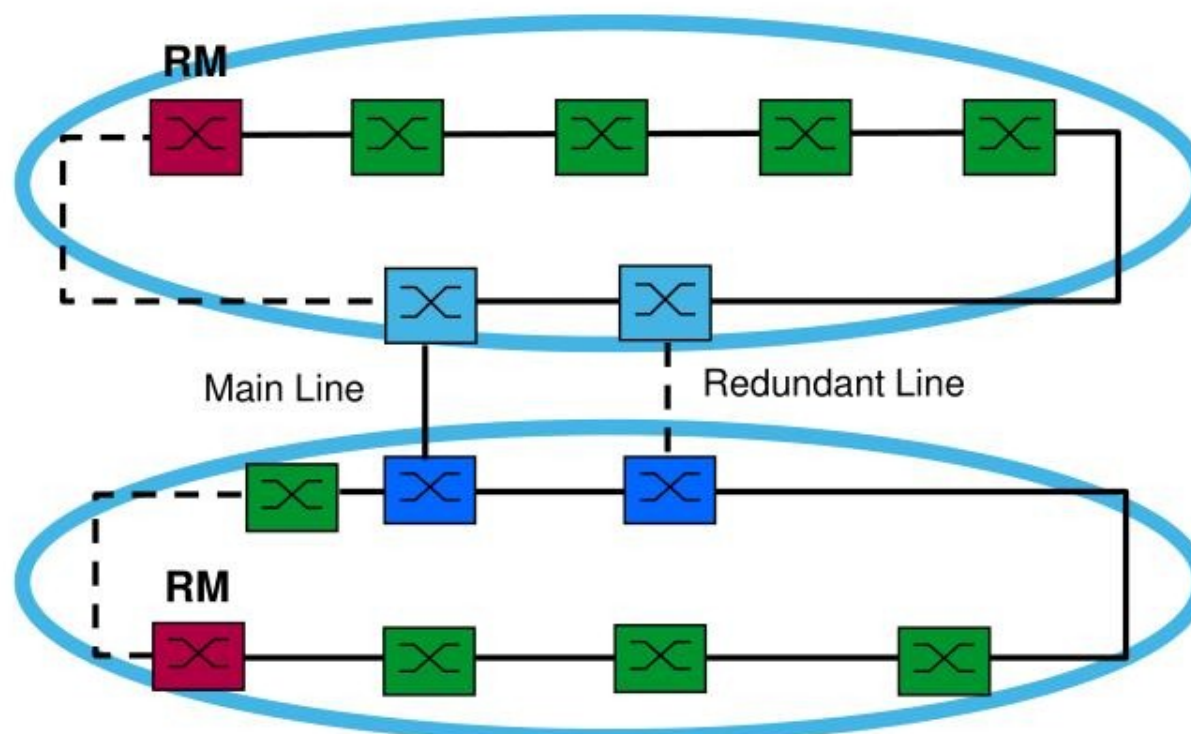
Refer to the ConneXium Switch Redundancy Manual for other Ring/Network Coupling Configurations

Redundancy Capabilities

Ring/Network Coupling

- Two switch coupling network configuration

- Ring coupling detects main line inoperable, the redundant switch resumes the communication via the redundant line typically within 150 ms (max. 500 ms)



Refer to the ConneXium Switch Redundancy Manual for other Ring/Network Coupling Configurations

Redundancy Capabilities

Ring/Network Coupling

- This web page allows the selection of:
 - Ring/Network coupling configurations.
 - Port Selections
 - Redundancy Modes
 - Coupling Modes

The screenshot shows a web interface for configuring Ring/Network Coupling. The title bar is green and says "Ring/Network Coupling".

Select Configuration: Five diagrams show different coupling topologies between two modules. The first diagram is selected, showing a direct connection between the two modules.

Select Port: Fields for selecting ports and modes.

[Module . Port]		Port mode	Port state
Coupling port	1 . 4	active	not connected
Partner coupling port		stand-by	not connected
IP Address		0.0.0.0	
Control port	1 . 0		not connected

Operation: ☒ On, ☐ Off

Information: ☐ Redundancy existing, ☐ Configuration failure

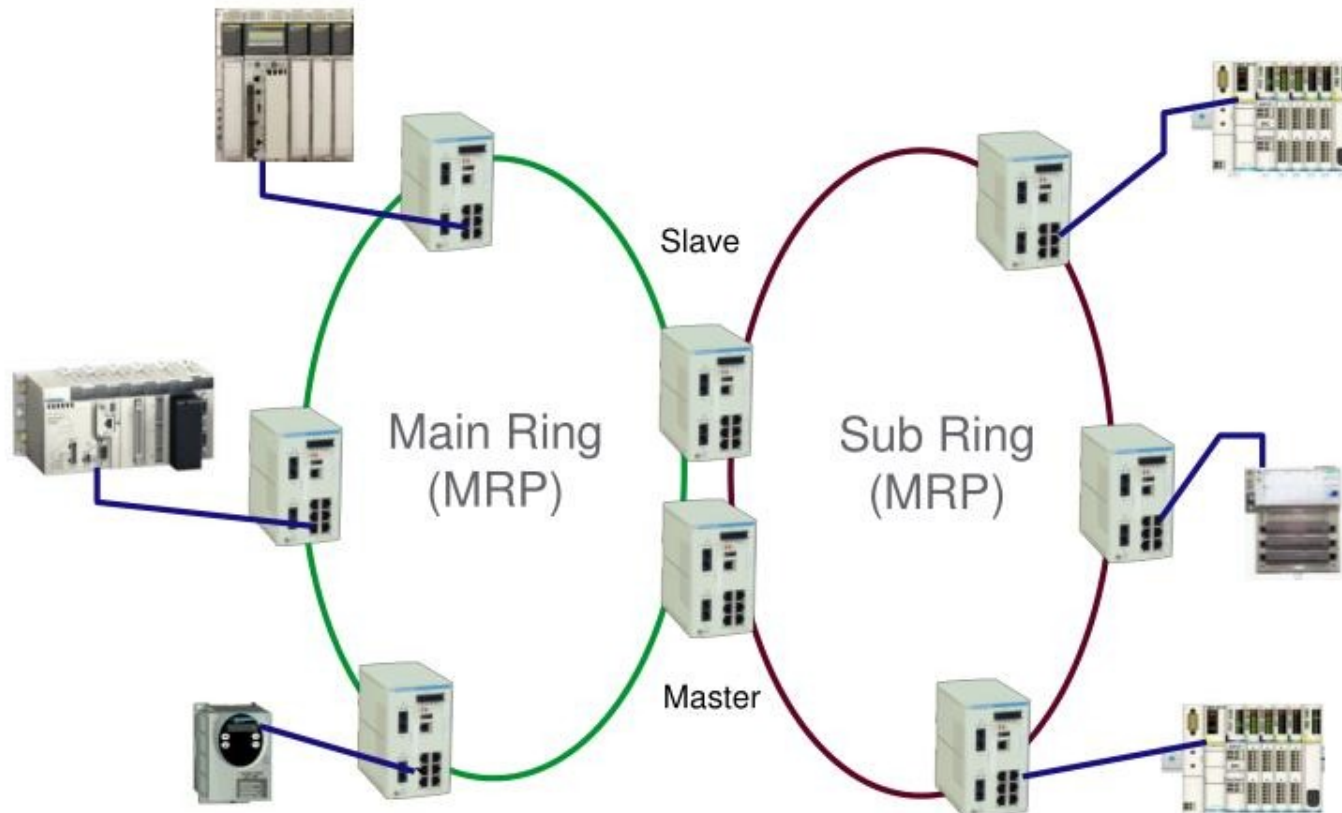
Redundancy Mode: ☐ Redundant Ring/Network Coupling, ☒ Extended Redundancy

Coupling Mode: ☒ Ring Coupling, ☐ Network Coupling

Redundancy

Dual Ring Architecture

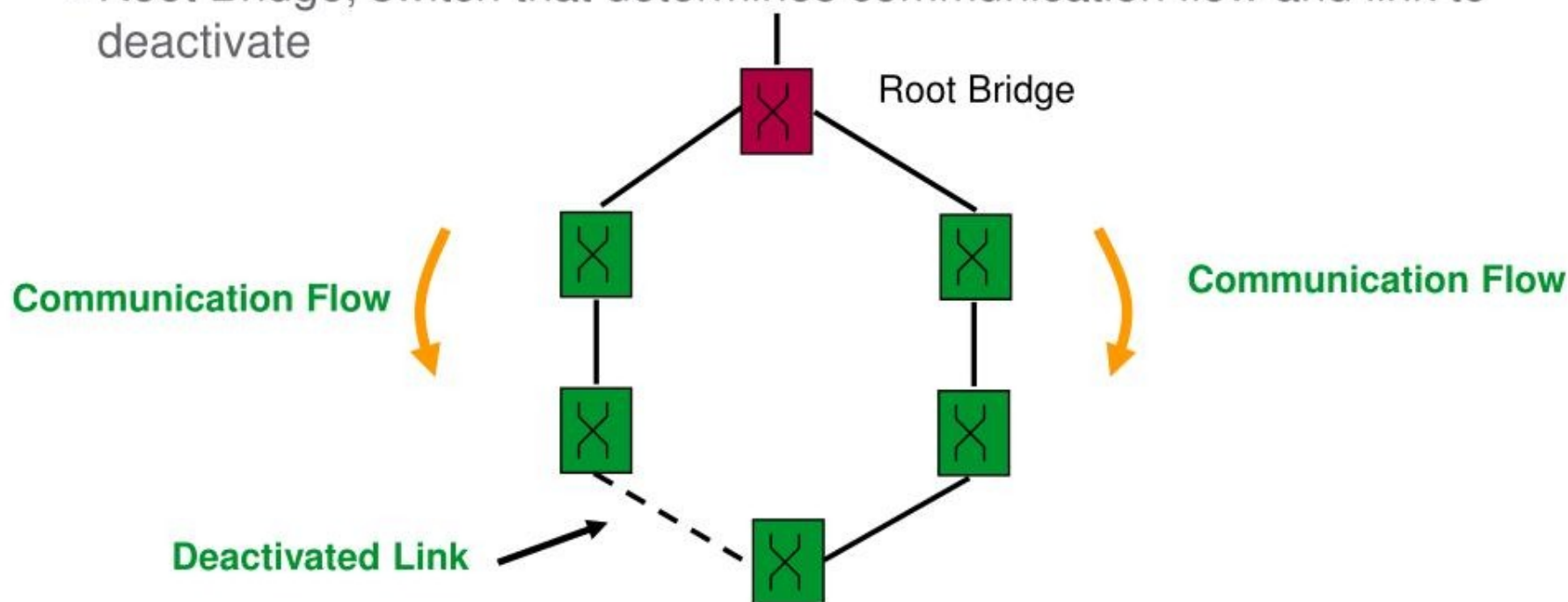
- Dual Rings with Main Ring and Sub Rings connected by dual switches provide redundant connection between rings



Redundancy Capabilities

RSTP (Rapid Spanning Tree Protocol)

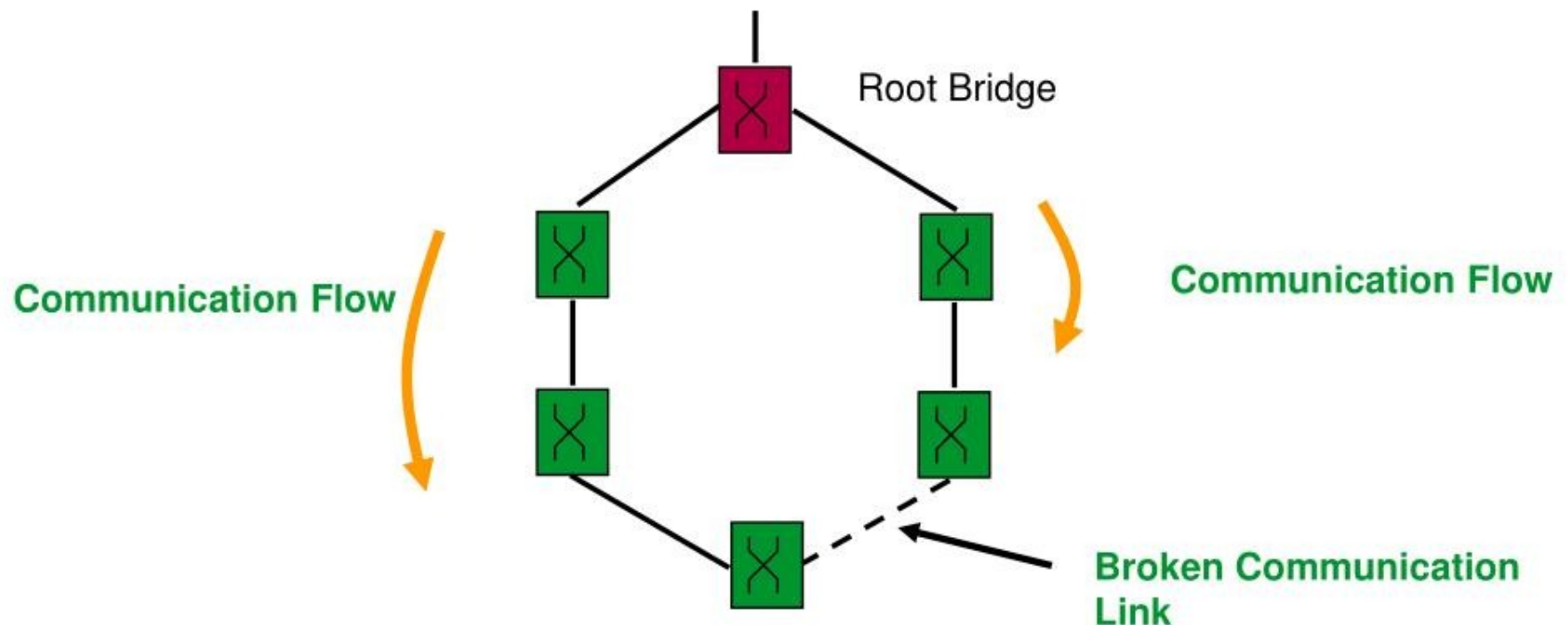
- A multiple connection network architecture that is used in a ring configuration
- On loss of a connection or switch, RSTP will reactivate a previously deactivated connection to maintain communications
- Root Bridge, switch that determines communication flow and link to deactivate



Redundancy Capabilities

RSTP (Rapid Spanning Tree Protocol)

- On loss of a communication link or switch, communication flow is automatically reconfigured and the deactivated link is reactivated to maintain communications.



Redundancy Capabilities

RSTP (Rapid Spanning Tree Protocol)

- This web page allows the selection of the RSTP capability
 - MRP compatibility, maintains the fast switching times of MRP with RSTP
 - Priority, sets local bridge priority combined with the switch MAC address are the bridge identification.
 - Hello Time, in sec. between the sending of two configuration messages
 - Forward Delay, transition time between changing states (forwarding, blocking)
 - Max Age, number of devices (max. branch length) to the root bridge

The screenshot shows the 'RSTP Global' configuration page. It features a green header with the title 'RSTP Global'. Below the header, there are two sections: 'Operation' and 'MRP compatibility'. The 'Operation' section has radio buttons for 'On' (selected) and 'Off'. The 'MRP compatibility' section has radio buttons for 'On' and 'Off' (selected). Below these is the 'Root Information' section, which includes fields for 'Root Id' (32768), 'Root Port' (0.0), and 'Root Cost' (0). There is also a 'Priority' field (32768) and a 'MAC Address' field (00:80:63:3c:22:34). A checkbox labeled 'This device is root' is checked. Below the 'Root Information' section is the 'Protocol Configuration/Information' section, which includes fields for 'Priority' (32768), 'MAC Address' (00:80:63:3c:22:34), 'Hello Time [s]' (2), 'Forward Delay [s]' (15), 'Max Age [s]' (20), 'Topology Changes' (0), and 'Time since last change' (7 day(s), 7:35:41). At the bottom is an 'Information' section.

Operation		MRP compatibility	
☒ On	☐ Off	☐ On	☒ Off

Root Information	
Root Id	32768
Root Port	0.0
Root Cost	0
Priority	32768
MAC Address	00:80:63:3c:22:34
☒ This device is root	

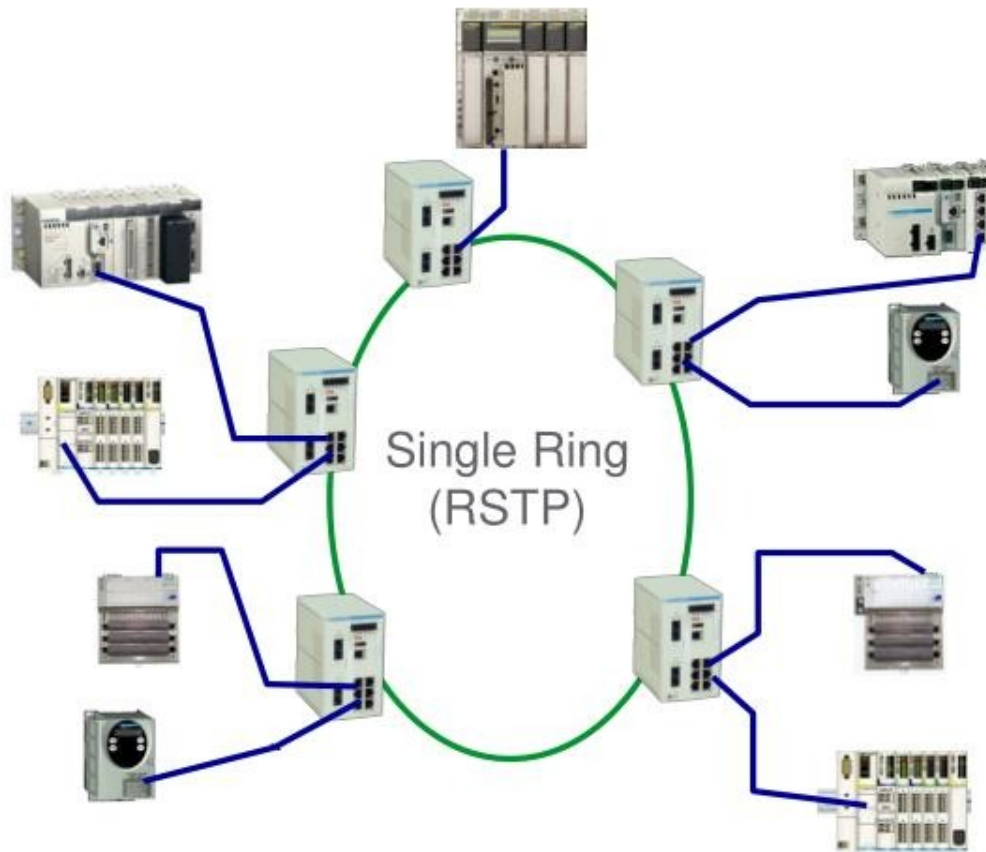
Protocol Configuration/Information	
Priority	32768
MAC Address	00:80:63:3c:22:34
Hello Time [s]	2
Forward Delay [s]	15
Max Age [s]	20
Topology Changes	0
Time since last change	7 day(s), 7:35:41

Information	
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Redundancy

RSTP (Rapid Spanning Tree Protocol)

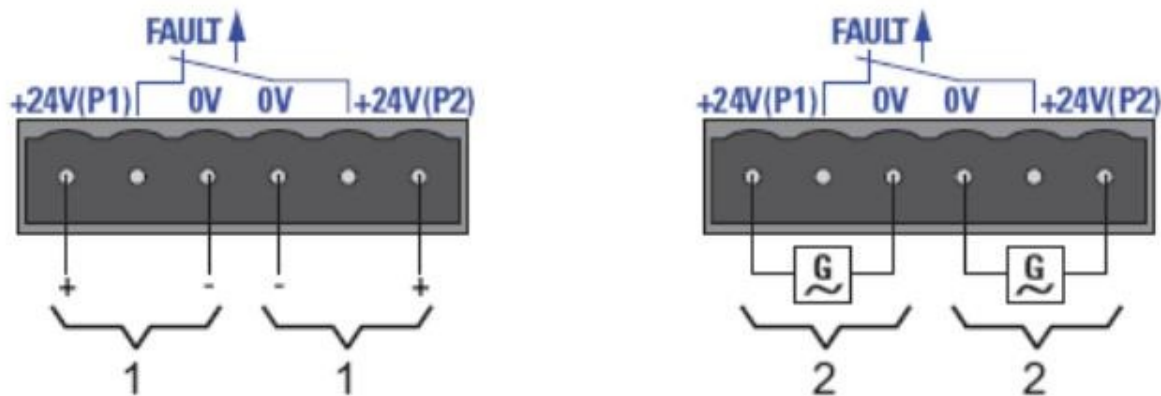
- Single Ring network, with connected devices, using RSTP



Redundancy

Dual Power Supply Connections

- Dual Power Source Connections for the Power Supply, allows the use of 2 separate power sources for redundancy.



Connecting the supply voltage at the 6-pin terminal block
1 – DC voltage, voltage range: 9.6 V DC to 60 V DC
2 – AC voltage, voltage range: 18 V AC to 30 V AC

Make the most of your energy